



TURNKEY PROJECT | CASE STUDY

# Cloud-Managed LAN & WiFi 6 Network

Petchem Office • Dahej LNG Terminal

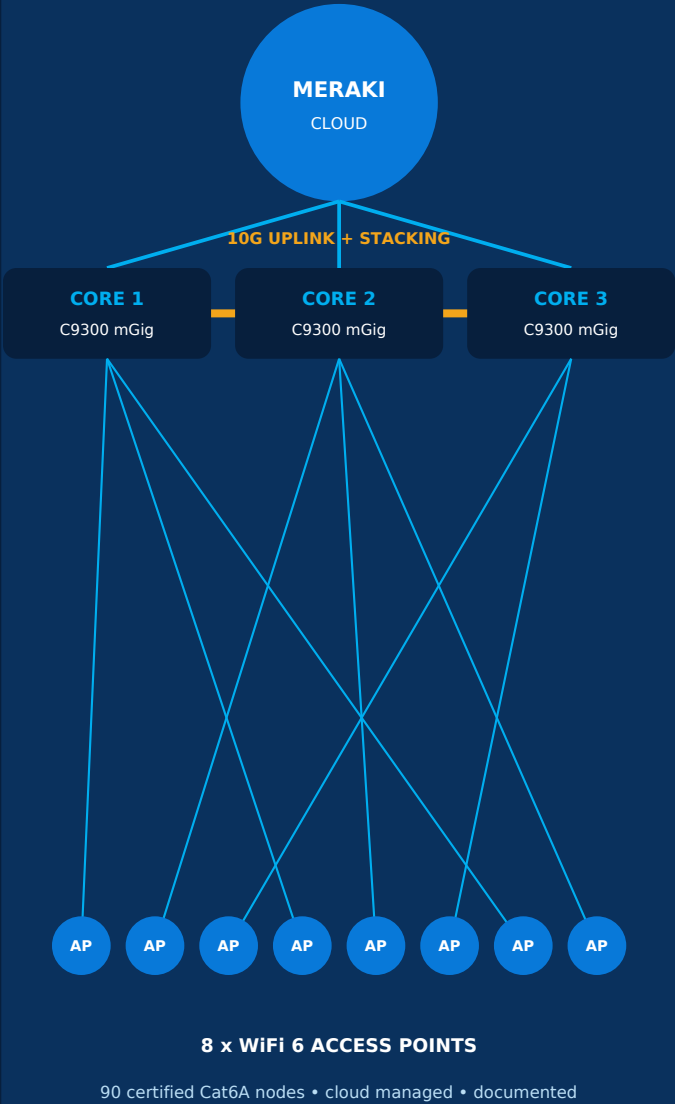
CORE • FIBRE • WIFI • CAT6A

## PROJECT PROFILE

Client: Petronet LNG Limited  
Location: Petchem Office, Dahej LNG Terminal  
Project scale: Approx. USD 0.04M

## D-TECH SOLUTION INTEGRATORS PVT. LTD.

Lead with Trust. Inspire with Vision.



## PROJECT AT A GLANCE

# A modern digital foundation for the Petchem Office

Petronet LNG Limited selected D-TECH to implement a comprehensive Local Area Network at its Petchem Office in Dahej. The awarded solution combines Cisco Meraki cloud-managed switching, WiFi 6, 10G fibre uplinks, Cat6A structured cabling, rack infrastructure, testing, documentation and support within one coordinated scope.

**USD 0.04M**

Approx. project scale

**3**

Core switches

**8**

WiFi 6 APs

**90**

Certified nodes

## Business requirement

- Deliver multi-gigabit wired access and high-capacity wireless coverage for a modern office environment.
- Create a resilient core with switch stacking, high-speed uplinks and managed power delivery for wireless access points.
- Provide cloud-based configuration and monitoring while maintaining clear security policies, SSIDs and operational control.
- Establish a labelled, certified and documented physical layer that supports troubleshooting, growth and handover.

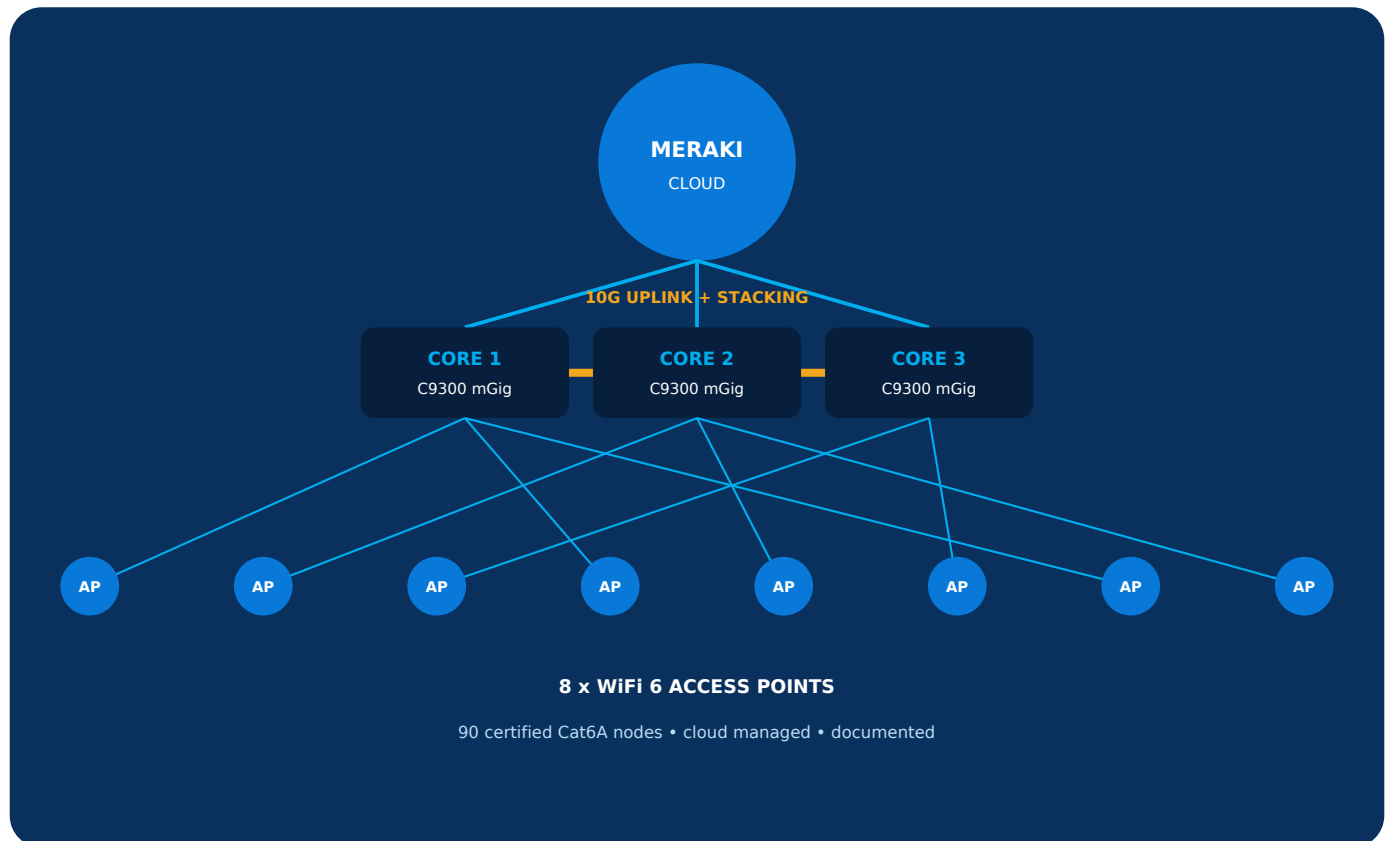
### D-TECH VALUE PROPOSITION

One accountable partner for survey, design alignment, materials, installation, configuration, certification, documentation, training and early-life support.

## CLOUD-MANAGED, HIGH-SPEED AND STRUCTURED

# An integrated Cisco Meraki network design

The architecture connects the cloud management plane to a three-switch core, 10G fibre uplinks, eight WiFi 6 access points and a professionally terminated Cat6A access layer.



## Architecture value

### VISIBILITY

Meraki cloud configuration and monitoring centralise operational control.

### CAPACITY

Multi-gigabit switching, PoE+ and WiFi 6 support modern device density.

### RESILIENCE

Stacking, redundant power design and 10G uplinks strengthen continuity.

### SECURITY

Defined network policies, SSIDs and WPA3-capable wireless improve governance.

## ACTIVE AND PASSIVE INFRASTRUCTURE

# A complete equipment and installation package

The final composite order and detailed price schedule define the following key quantities. Values below are taken directly from the awarded scope.

| Active network component                      | Qty   | Purpose                              |
|-----------------------------------------------|-------|--------------------------------------|
| Cisco C9300-24UX-M Meraki-enabled core switch | 3     | 24-port multi-gigabit UPOE switching |
| C9300-NM-8X-M network module                  | 3     | 8 x 10GE uplink connectivity         |
| 1100W AC Platinum power supply                | 3     | Power and redundancy setup           |
| 50 cm stacking cable                          | 3     | Switch stack connectivity            |
| 10G Base-LR single-mode SFP+                  | 2     | High-speed fibre backbone            |
| Meraki MR44 WiFi 6 indoor access point        | 8     | Indoor wireless coverage             |
| Five-year Meraki enterprise licences          | 3 + 8 | Switch and access-point management   |

## Structured cabling and rack infrastructure

| Passive / service component                        | Verified quantity          |
|----------------------------------------------------|----------------------------|
| Cat6A UTP cable                                    | 12 boxes x 305 m = 3,660 m |
| Cat6A unloaded patch panels                        | 3                          |
| Cat6A modular jacks                                | 180                        |
| Single-port shuttered faceplates / certified nodes | 90                         |
| Cat6A rack patch cords                             | 100                        |
| 24U network rack with PDU and cable management     | 1 complete set             |
| PVC conduit and casing-capping installation        | 1,500 m                    |
| Cable laying, labelling, termination and testing   | Complete contracted scope  |

## END-TO-END TECHNOLOGY EXECUTION

# Eight clearly defined delivery points

D-TECH's scope combines network engineering, physical infrastructure, cloud configuration and lifecycle handover. Each delivery point is connected to one coordinated project plan.

**SITE SURVEY**

Cable routes, access-point placement, compatibility and power assessment.

**CAT6A NETWORK**

3,660 m cabling, 90 nodes, termination, labelling and certification.

**RACK & PATHWAYS**

24U rack, PDUs, conduits, cable protection and professional dressing.

**CORE SWITCHING**

Three Cisco C9300 multi-gigabit UPOE switches with stacking.

**10G FIBRE**

Single-mode LR uplinks, fibre termination and backbone verification.

**WIFI 6 COVERAGE**

Eight Meraki MR44 access points with PoE+ and placement validation.

**CLOUD & SECURITY**

Meraki licensing, dashboard, policies, SSIDs and security setup.

**TEST & HANDOVER**

Connectivity tests, diagrams, backups, documentation and training.

## Assurance commitments

| Control area           | Contracted requirement                                                              |
|------------------------|-------------------------------------------------------------------------------------|
| Operational continuity | Minimise disruption and coordinate access and scheduling with PLL officers.         |
| Safety & quality       | Follow PLL safety rules, use genuine OEM products and follow manufacturer guidance. |
| Installation warranty  | Minimum one-year onsite warranty for installation work.                             |
| Early-life support     | 24/7 support during the initial 30 days after commissioning.                        |
| Documentation          | Network diagram, as-built records, configuration backup and knowledge transfer.     |
| Contract schedule      | Completion scheduled on or before 28 February 2026.                                 |

## A SCALABLE NETWORK OPERATING MODEL

# Value designed into the contracted solution

The order establishes the technology and delivery controls needed for a supportable office network. The benefits below describe design intent and contracted capability; no unverified uptime, ROI or completion claim has been added.

### CENTRALISED CONTROL

Cloud dashboard configuration gives the IT team a unified view of switches, access points, licences and policy.

### HIGHER PERFORMANCE

Multi-gigabit switching, 10G fibre and WiFi 6 provide a platform for data-intensive office services.

### WIRELESS READINESS

Eight optimally placed indoor access points support coverage, capacity and modern security options.

### SERVICEABILITY

Labelling, professional rack dressing, cable certification and as-built documentation improve support readiness.

### LIFECYCLE ASSURANCE

Five-year cloud licences, installation warranty and early-life support extend value beyond deployment.

### ONE-PARTNER DELIVERY

A single integrator coordinates active, passive, civil-support, testing, commissioning and handover workstreams.

### INTERNATIONAL PROJECT SCALE

Approx. USD 0.04 million. Rounded presentation equivalent for case-study communication; the original commercial contract remains denominated in INR.